EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 01

Basic Maths and Calculus

Trigonometry

By — Excellence Programme

1 September 2026

Topics to be covered

01 Trigonometric ratios — sin, cos, tan, cosec, sec, cot

02 Trigonometric identities from $P^2 + B^2 = H^2$

03 Finding unknown sides from a given ratio

04 Range of sine and cosine

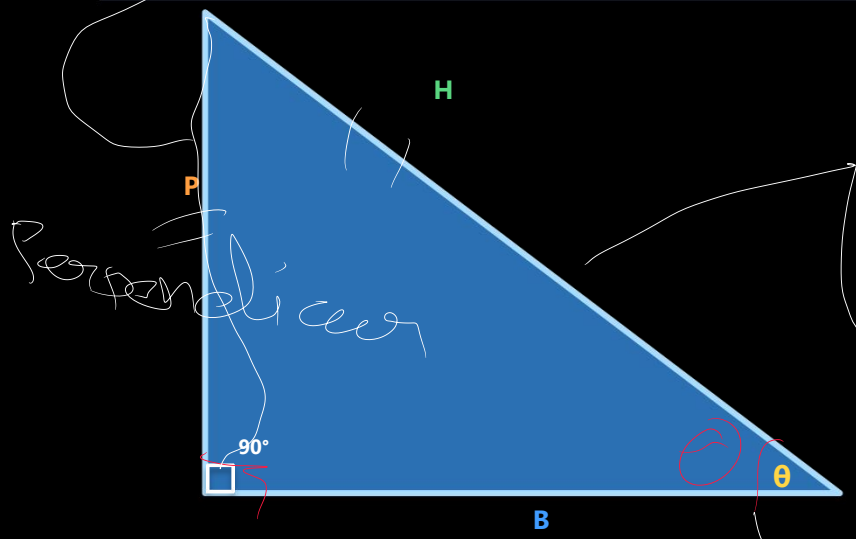
05 Algebra with ratios, and similar triangles

06 Height and distance

07 Standard angles 0° to 180°, and 37° / 53°

08 Radian measure, arc = Rθ, degree ↔ radian

Name the sides from θ



P — perpendicular — opposite θ
B — base — adjacent to θ
H — hypotenuse — longest side

Not from how the figure sits.

$$\sin\theta = \frac{P}{H}$$

$$\cos\theta = \frac{B}{H}$$

$$p^2 + b^2 = h^2$$

sin θ and cosec θ

sin θ = P / H

Opposite over hypotenuse.

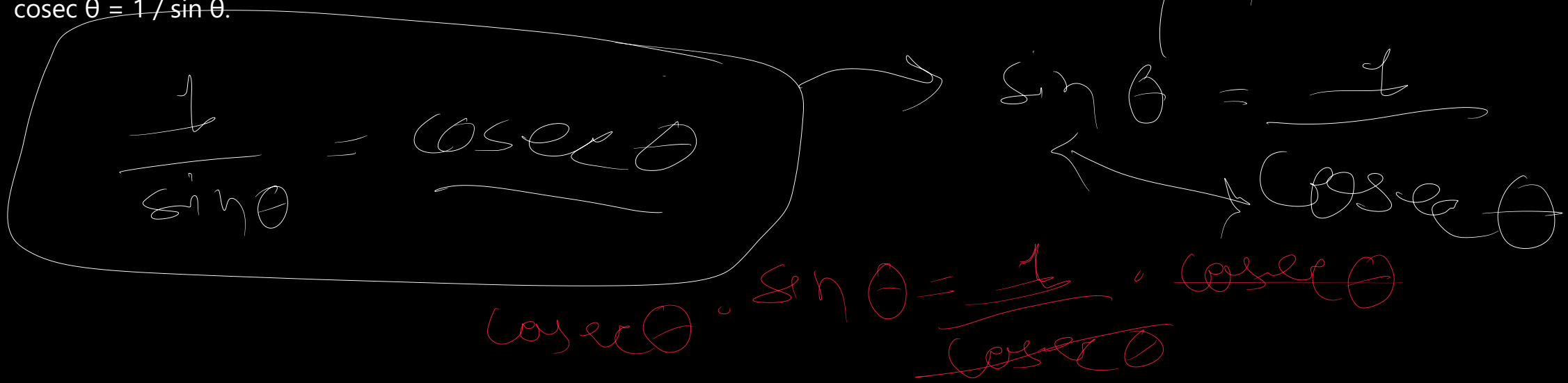
cosec θ = H / P

Reciprocal of sin θ. If sin θ = 3/5, then cosec θ = 5/3.

sin, cos, tan are dimensionless. Never write sin θ = 0.6 m.

cosec θ = 1 / sin θ.

sin θ · cosec θ = 1



cos θ and sec θ

cos θ = B / H

Adjacent over hypotenuse.

sec θ = H / B

Reciprocal of cos θ. If cos θ = 4/5, then sec θ = 5/4.

Never write cos θ = 0.8 m. A ratio has no unit.

sec θ = 1 / cos θ.

cos θ · sec θ = 1

$\cos\theta = \frac{1}{\sec\theta}$

$\cos\theta = \frac{B}{H} \rightarrow \sec\theta = \frac{H}{B} = \frac{1}{\frac{B}{H}}$

cos θ and sec θ

cos θ = B / H

Adjacent over hypotenuse.

sec θ = H / B

Reciprocal of cos θ. If cos θ = 4/5, then sec θ = 5/4.

Never write cos θ = 0.8 m. A ratio has no unit.

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$\cos\theta = \frac{1}{\sec\theta}$

$\cos\theta = \frac{B}{H} \rightarrow \sec\theta = \frac{H}{B} = \frac{1}{\frac{B}{H}}$

tan θ and cot θ

tan θ = P / B

Opposite over adjacent.

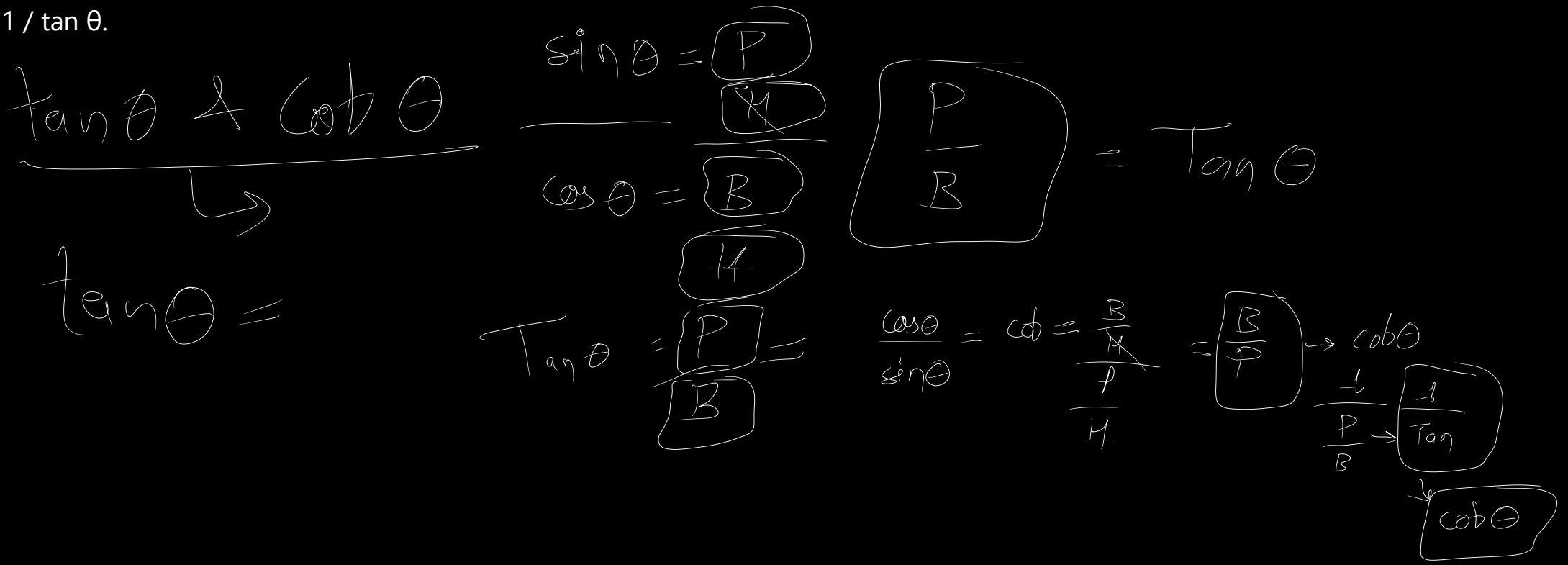
cot θ = B / P

Reciprocal of tan θ. Also: tan θ = sin θ / cos θ.

sin · cosec = 1 cos · sec = 1 tan · cot = 1

cot θ = 1 / tan θ.

tan θ · cot θ = 1



Divide Pythagoras by H^2

Start here. Divide both sides by H^2.

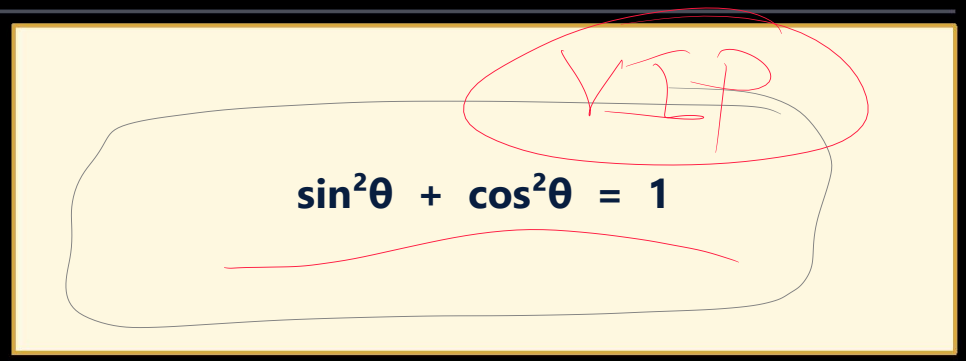
$$P^2 + B^2 = H^2$$

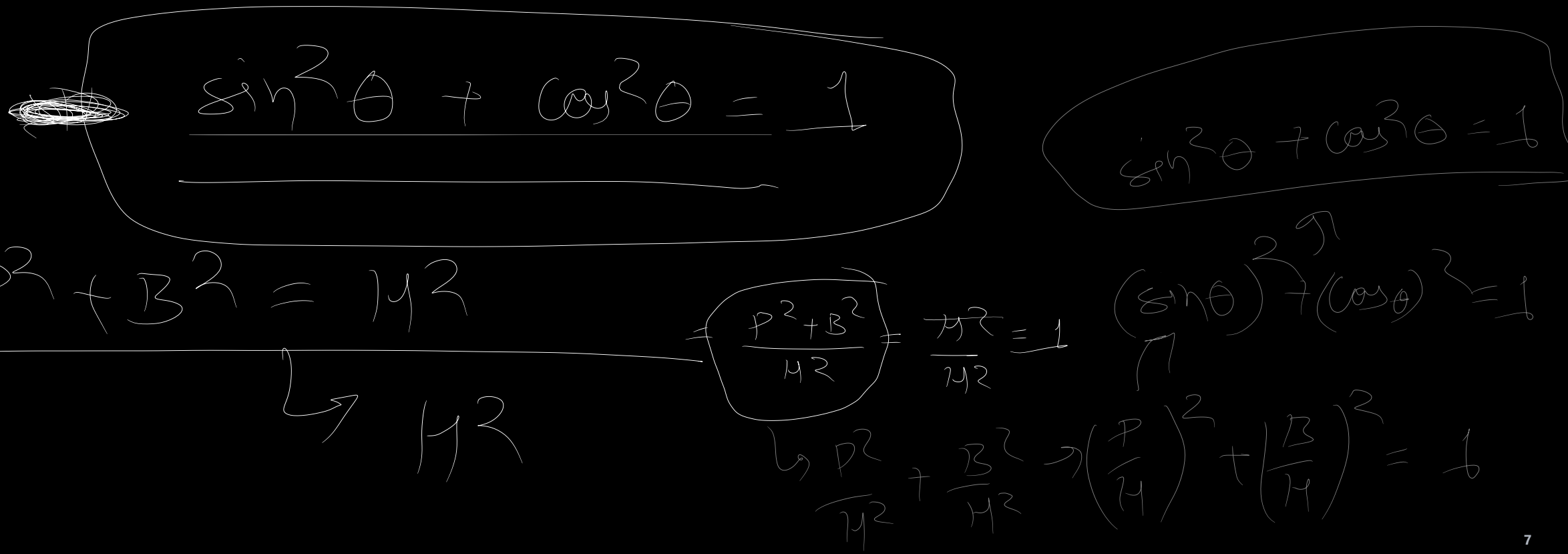
$$P^2 / H^2 + B^2 / H^2 = H^2 / H^2$$

$$(P / H)^2 + (B / H)^2 = 1$$

Also: $\cos\theta = \sqrt{(1 - \sin^2\theta)}$ and $\sin\theta = \sqrt{(1 - \cos^2\theta)}$

$$\sin^2\theta + \cos^2\theta = 1$$





Divide Pythagoras by H^2

Start here. Divide both sides by H^2.

$$P^2 + B^2 = H^2$$

$$P^2 / H^2 + B^2 / H^2 = H^2 / H^2$$

$$(P / H)^2 + (B / H)^2 = 1$$

Also: $\cos\theta = \sqrt{(1 - \sin^2\theta)}$ and $\sin\theta = \sqrt{(1 - \cos^2\theta)}$

$$\sin^2\theta = 1 - \cos^2\theta$$

$$\sin\theta = \sqrt{1 - \cos^2\theta}$$

$$\sin^2\theta + \cos^2\theta = 1$$

Divide Pythagoras by B^2

Same start. Now divide both sides by B^2.

$$P^2 + B^2 = H^2$$

$$P^2 / B^2 + B^2 / B^2 = H^2 / B^2$$

$$(P / B)^2 + 1 = (H / B)^2$$

Because $P/B = \tan\theta$ and $H/B = \sec\theta$.

$$1 + \tan^2\theta = \sec^2\theta$$

Divide Pythagoras by P^2

Same start. Now divide both sides by P^2.

$$P^2 + B^2 = H^2$$

$$P^2 / P^2 + B^2 / P^2 = H^2 / P^2$$

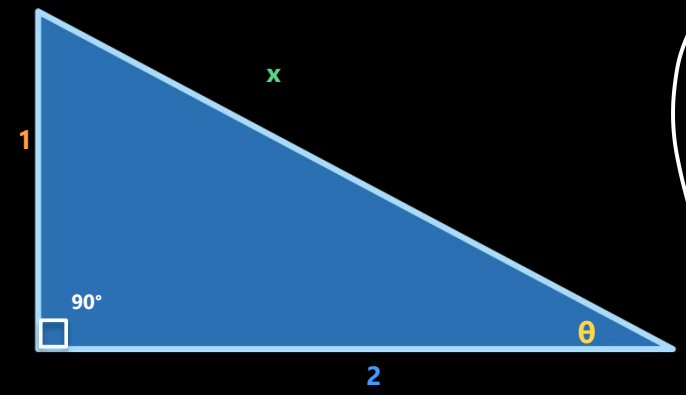
$$1 + (B / P)^2 = (H / P)^2$$

Because $B/P = \cot\theta$ and $H/P = \operatorname{cosec}\theta$. If you keep only one line: $\sin^2\theta + \cos^2\theta = 1$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

Find x

In the right triangle shown, the perpendicular is 1 and the base is 2. The hypotenuse is x. Find the value of x, sin θ, cos θ and tan θ.



P = 1

B = 2

H = ? = x

$P^2 + B^2 = H^2$

$1^2 + 2^2 = H^2$

$1 + 4 = H^2 \rightarrow$

$H^2 = 5$

$H = \sqrt{5}$

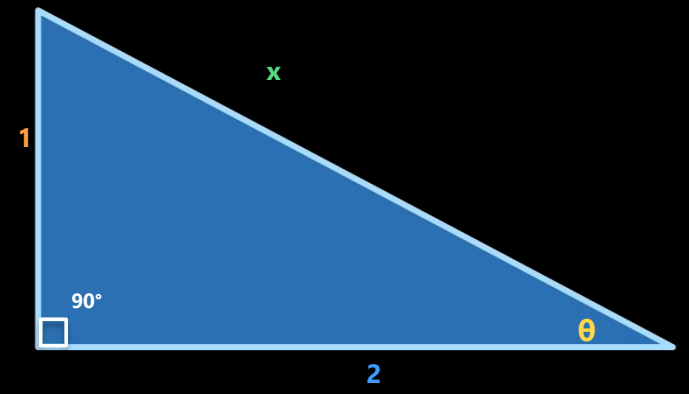
$\sin\theta = \frac{1}{\sqrt{5}}$

$\cos\theta = ?$

$= \frac{2}{\sqrt{5}}$

First find H

Right triangle: P = 1, B = 2, H = x.



$P^2 + B^2 = H^2$

$1^2 + 2^2 = x^2$

$x = \sqrt{5}$

Now sin θ, cos θ, tan θ

P = 1, B = 2, H = √5

sin θ = P / H = 1 / √5
cos θ = B / H = 2 / √5
tan θ = P / B = 1 / 2

Check: (1/√5) / (2/√5) = 1/2. Same.

tan θ = sin θ / cos θ = 1/2

$$\tan = \frac{P}{B} = \frac{1}{2}$$

If tan θ = 2

$P = x$

If tan θ = 2, then find the perpendicular, the base and the hypotenuse. Also find sin θ and cos θ. Take θ acute.

$\tan = 2 = \frac{P}{B} \rightarrow \frac{x}{B} = 2$

$x = 2B$

$x = 1 \quad B = 2$

$x = 2 \rightarrow B = 4$

$P = 4 \rightarrow B = 8$

$H^2 = P^2 + B^2 = 4^2 + 8^2 = 16 + 64 \Rightarrow 80$

$H^2 = 80 \rightarrow$

$H = \sqrt{80}$

$H = \sqrt{2 \times 2 \times 2 \times 2 \times 5}$

$2 \times 2 \times \sqrt{5}$

$= 4\sqrt{5}$

$\sin\theta = ? = \frac{P}{H} \rightarrow$

$= \frac{4}{\sqrt{\quad}}$

$\sin\theta = \frac{x}{4\sqrt{5}} = \frac{1}{\sqrt{5}}$

$P = 1 \rightarrow \frac{1}{\sqrt{3}}$

$P = 1 \rightarrow B = 2 \rightarrow H^2 = 1 + 4 = 5$

$\hookrightarrow \sqrt{5} = H$

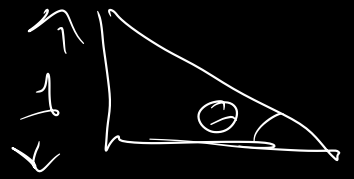
$H = \sqrt{5}$

$\sin\theta = \frac{1}{\sqrt{5}}$

If tan θ = 2

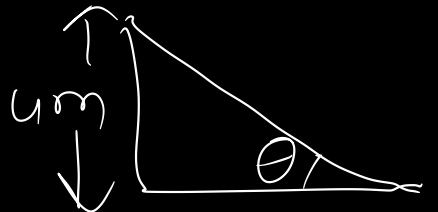
If tan θ = 2, then find the perpendicular, the base and the hypotenuse. Also find sin θ and cos θ. Take θ acute.

P = 1 →



$\sin\theta = \frac{1}{\sqrt{5}}$

P = 4



$\sin\theta = \frac{4}{\sqrt{5}}$

If θ is same ratio side of 2 Triangle is always Equal

SOLUTION

$\tan\theta = 2 \rightarrow$ find P, B, H

Given $\tan\theta = 2$ (θ acute).

$$\tan\theta = P/B = 2 = 2/1$$

Let B = k. Then P = 2k.

$$H^2 = (2k)^2 + k^2 = 5k^2$$

$H = \sqrt{5}\,k$ (take + because θ is acute)

P = 2k B = k H = √5 k

Now sin θ and cos θ

P = 2k, B = k, H = √5 k

sin θ = P / H = 2k / √5 k = 2 / √5

k cancels.

cos θ = B / H = k / √5 k = 1 / √5

k cancels.

A ratio only fixes the shape. The scale k always cancels in sin, cos, tan.

Range of sin θ

If sin θ = 2, then find the value of cos θ and tan θ.

sin θ → Range of sin θ is always in between

−1 & 1

$-1 \le \sin\theta, \cos\theta \le 1$

VIP

$\frac{\sqrt{-12}}{2} \rightarrow \frac{\sqrt{-3}}{\sqrt{}}$

Method 1 — geometry

If sin θ = 2, find cos θ and tan θ.

Wrong: Start calculating cos θ from the identity without checking the range.

Right: First ask: can sin θ equal 2?

→ first check Range

sin θ = P / H

H is the longest side, so H > P. Then P/H ≤ 1.

sin θ = 2 is impossible. There is no such triangle.



$\sin\theta = 2 \Rightarrow \frac{P}{H}$ → $P = 2H$

P can't be twice of H

→ H is longest side

Method 2 — identity



Try the identity anyway. It also fails.

$\sin^2\theta + \cos^2\theta = 1$

Put $\sin\theta = 2$.

$(2)^2 + \cos^2\theta = 1$

$\cos^2\theta = 1 - 4 = -3$. Not a real value.

$-1 \le \sin\theta \le 1$ and $-1 \le \cos\theta \le 1$ $\tan\theta$ has no finite range.

Find tan θ from a sin–cos fraction

M. W

If (sin θ + cos θ) / (sin θ − cos θ) = 11/5 , then find the value of tan θ.

$$\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{11}{5}$$

$\Rightarrow 5(\sin\theta + \cos\theta) = 11(\sin\theta - \cos\theta)$

$= 5\sin\theta + 5\cos\theta = 11\sin\theta - 11\cos\theta$

$5\cos\theta + 11\cos\theta = 11\sin\theta - 5\sin\theta$

$16\cos\theta = 6\sin\theta$

$\frac{16}{6} = \frac{\sin\theta}{\cos\theta} =$

$\frac{8}{3} = \tan\theta$

$\tan\theta = ?$

$\tan\theta \Rightarrow \frac{8}{3}$

Cross multiply, then expand

$(\sin\theta + \cos\theta) / (\sin\theta - \cos\theta) = 11/5$

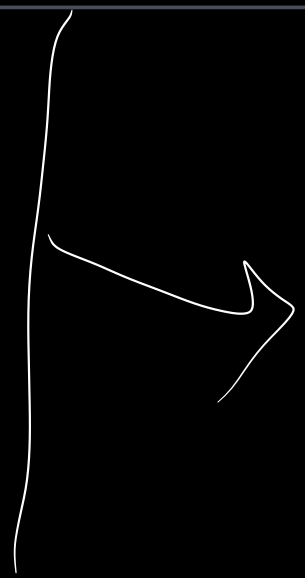
$5(\sin\theta + \cos\theta) = 11(\sin\theta - \cos\theta)$

Cross multiply.

$5\sin\theta + 5\cos\theta = 11\sin\theta - 11\cos\theta$

Open the brackets.

Next: collect sin on one side, cos on the other.



Collect, then divide

$5 \sin \theta + 5 \cos \theta = 11 \sin \theta - 11 \cos \theta$

$5 \cos \theta + 11 \cos \theta = 11 \sin \theta - 5 \sin \theta$

$16 \cos \theta = 6 \sin \theta$

$\sin \theta / \cos \theta = 16 / 6 = 8 / 3$

$\tan \theta = 8 / 3$

Nested right triangles — same θ, same tan θ

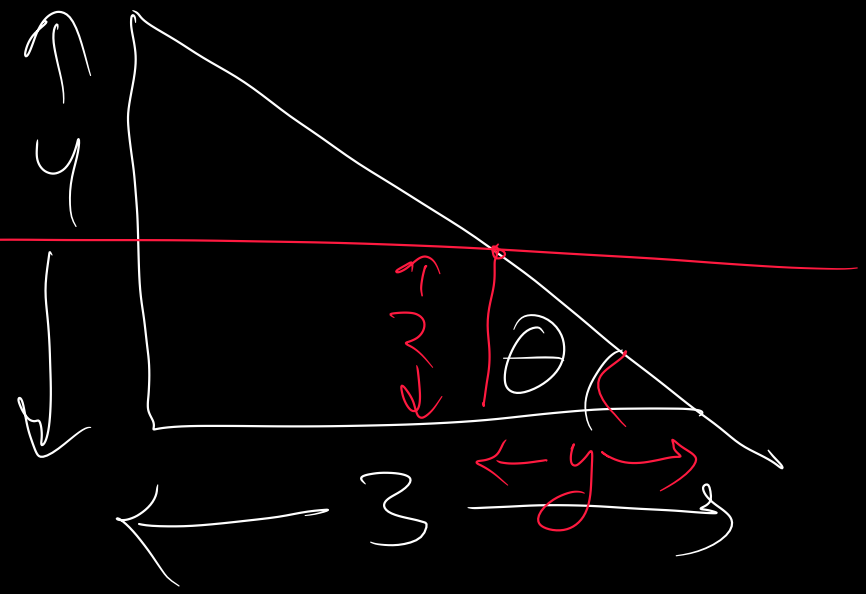
A large right triangle has opposite side 4 and adjacent side 3. Nested similar slices share the same angle θ. In the small slice the opposite is 2 and the adjacent is y. In the middle slice the opposite is x and the adjacent is 2. Find the values of x and y.

1. x = 4/3 , y = 3/2
2. x = 8/3 , y = 6/2
3. x = 8/3 , y = 3/2
4. x = 3/2 , y = 8/3

$\tan\theta = \frac{4}{3}$

$y = ?$

$\tan = \frac{P}{B} = \frac{4}{3}$



$\tan\theta = \frac{4}{3} = \frac{P}{B} = \frac{2}{y} \Rightarrow \frac{2}{y} = \frac{4}{3} \Rightarrow y = \frac{6}{4} = \frac{3}{2}$

$y = \frac{3}{2}$

Nested right triangles — same θ, same tan θ

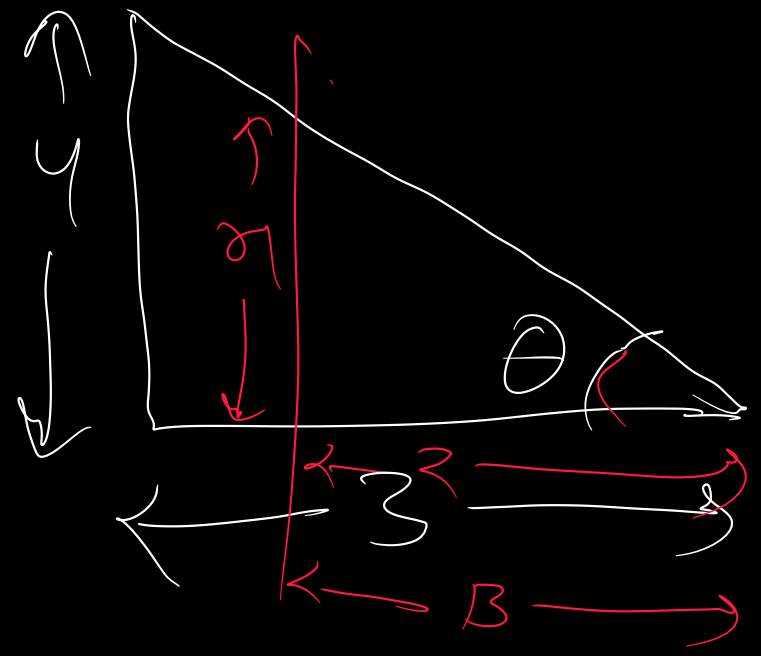
3

A large right triangle has opposite side 4 and adjacent side 3. Nested similar slices share the same angle θ. In the small slice the opposite is 2 and the adjacent is y. In the middle slice the opposite is x and the adjacent is 2. Find the values of x and y.

1. x = 4/3 , y = 3/2
2. x = 8/3 , y = 6/2
3. x = 8/3 , y = 3/2
4. x = 3/2 , y = 8/3

$x = ?$

$\tan\theta = \frac{4}{3} = \frac{x}{2}$



$x = \frac{8}{3}$ → $y = \frac{3}{2}$

for all nested △ $\tan\theta$, $\sin\theta$ & $\cos\theta$ is same

Find y from tan θ = 4/3

Large triangle: opposite 4, adjacent 3. Small slice: opposite 2, adjacent y.

tan θ = 4 / 3 = 2 / y

Same θ, so same tan.

4 y = 6

Cross multiply.

y = 3 / 2

Find x from tan θ = 4/3

Middle slice: opposite x, adjacent 2. Same θ as the large triangle.

tan θ = 4 / 3 = x / 2

Same θ, so same tan.

3 x = 8

Cross multiply.

x = 8 / 3

Do not mix a height from one triangle with a base from another. Each ratio must come from the SAME triangle.